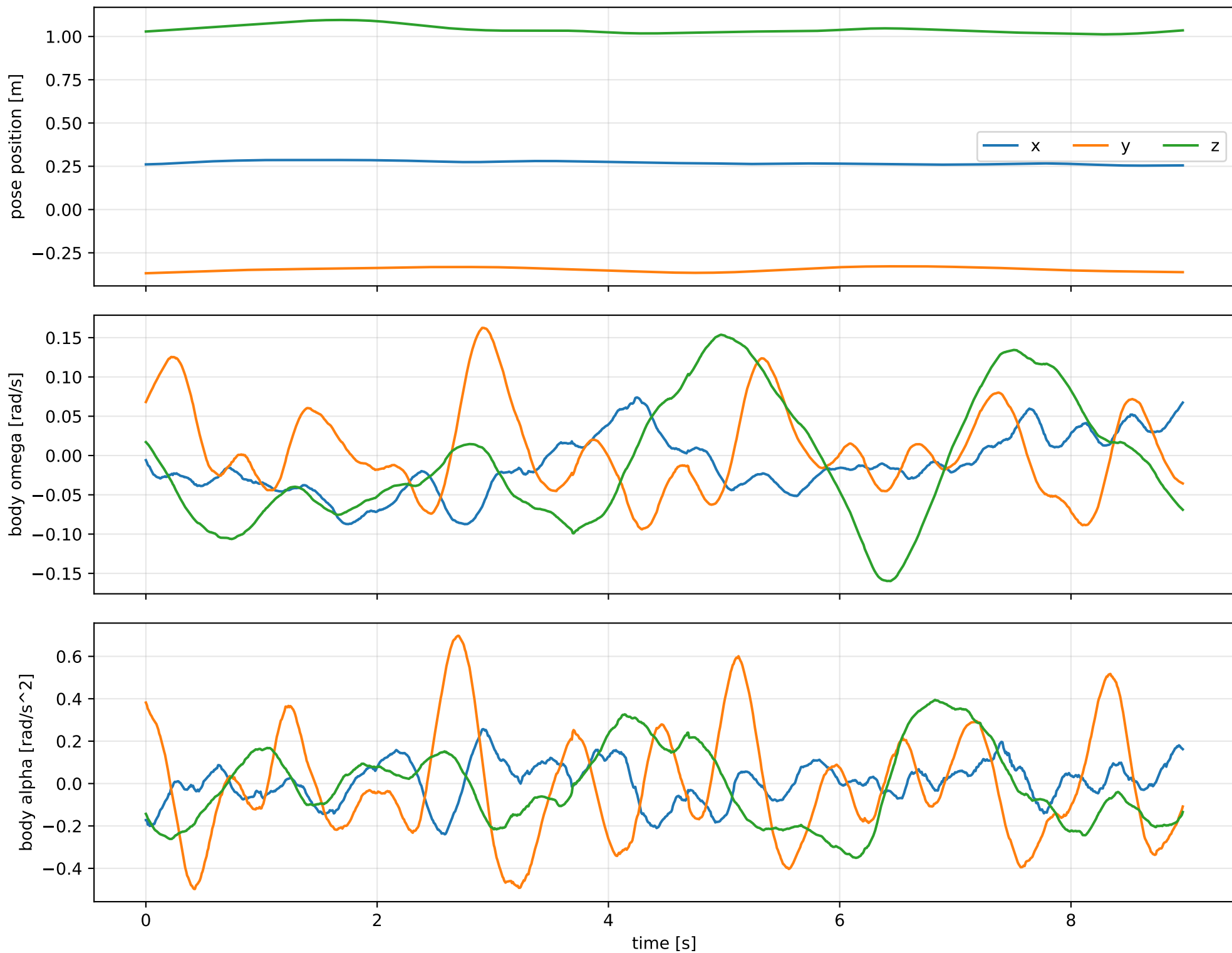
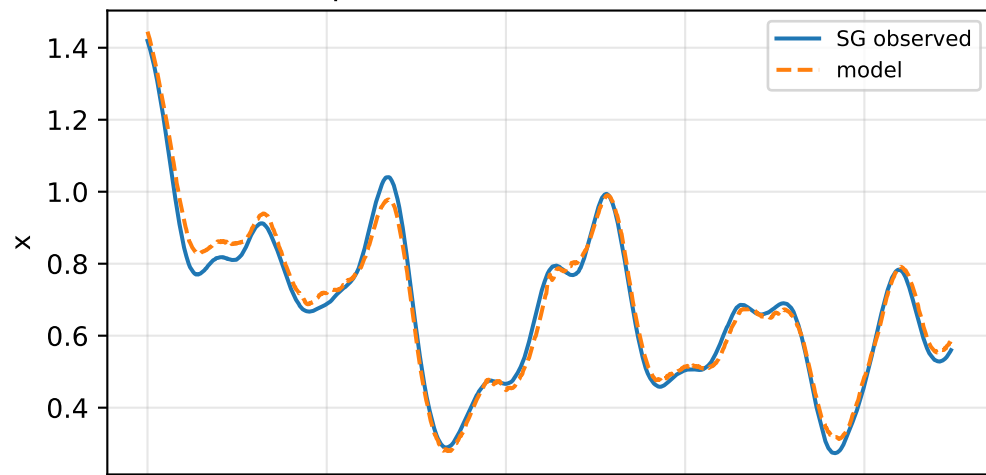


cov_full_no_R_uncertainty_in_s: SG trajectory and derived motion

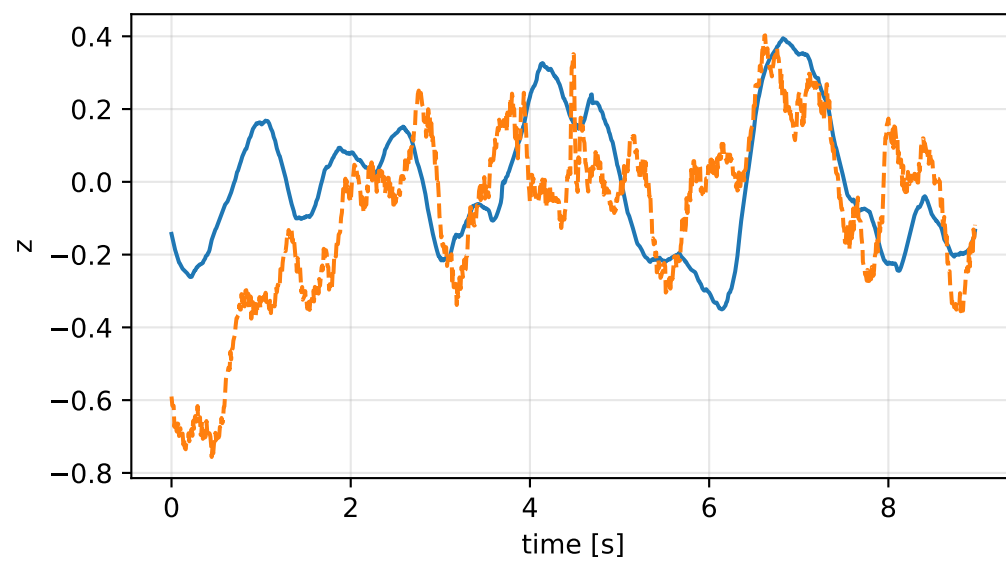
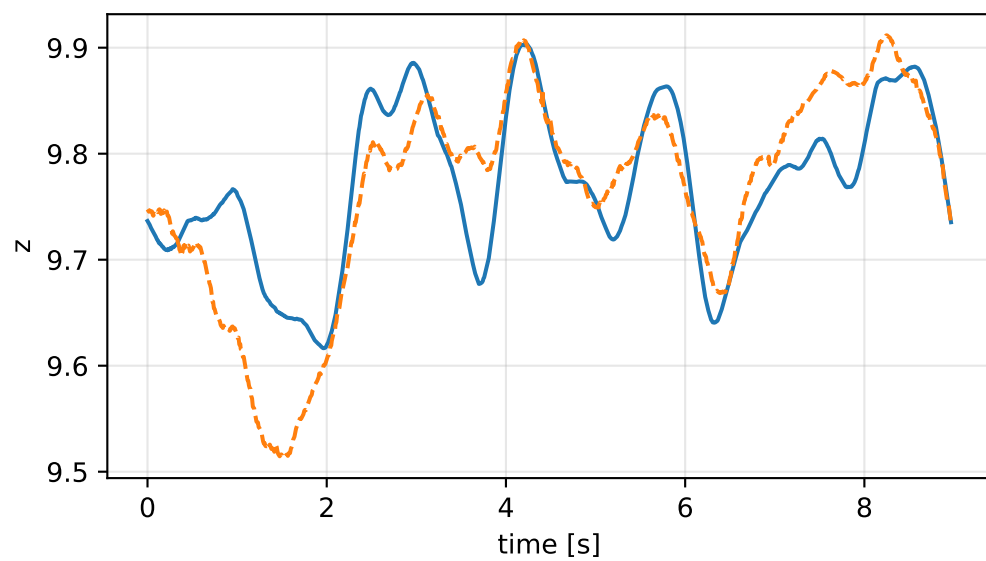
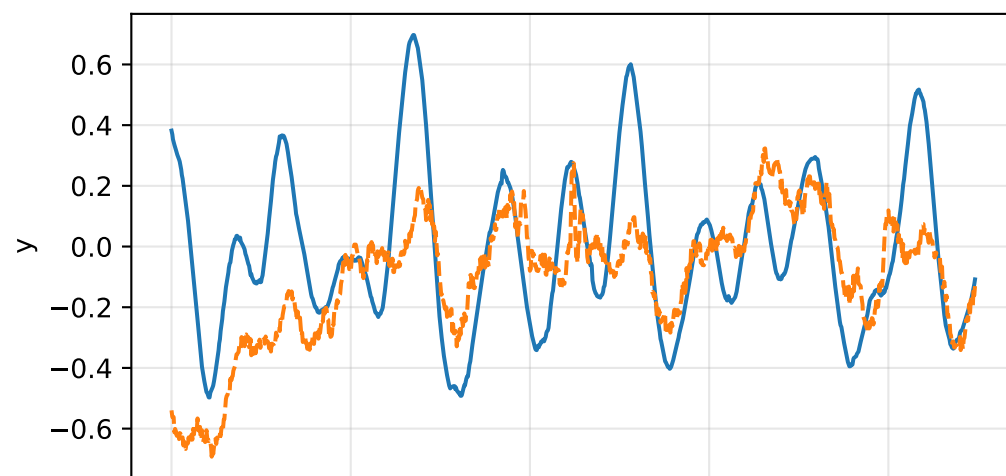
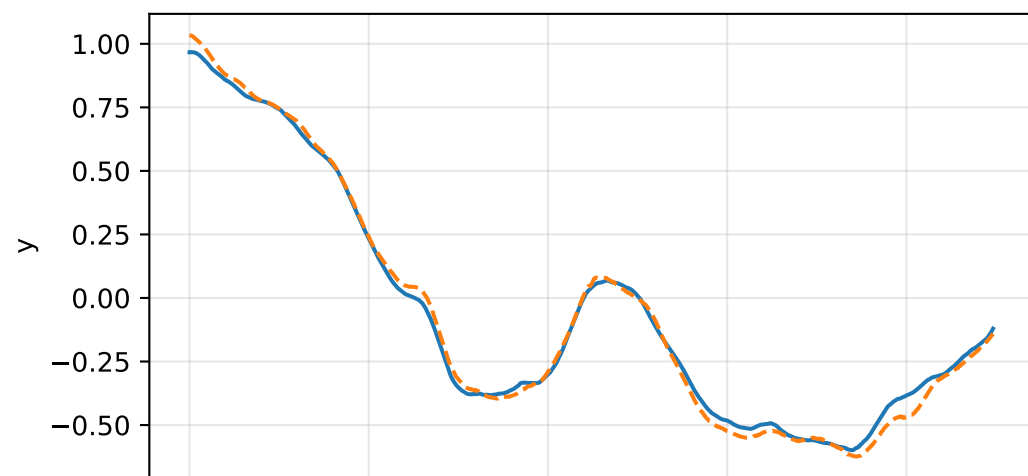
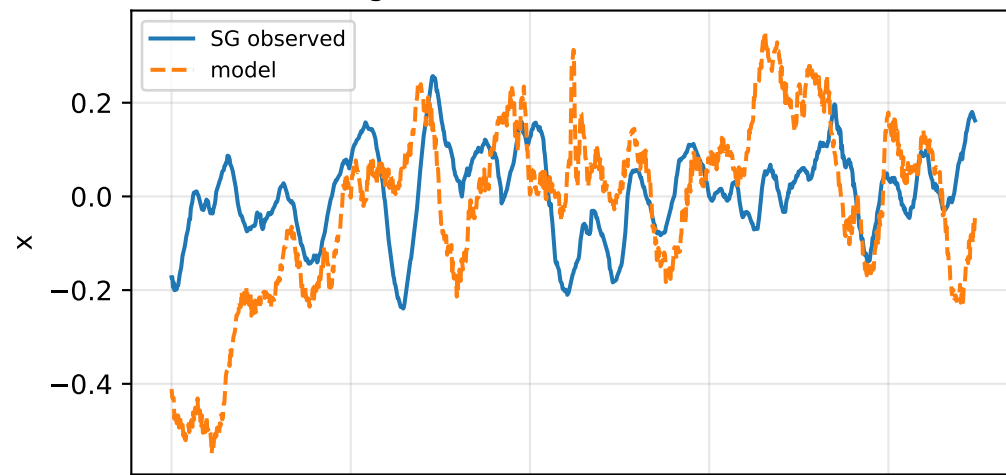


cov_full_no_R_uncertainty_in_s: acceleration residual objective

specific acceleration [m/s²]

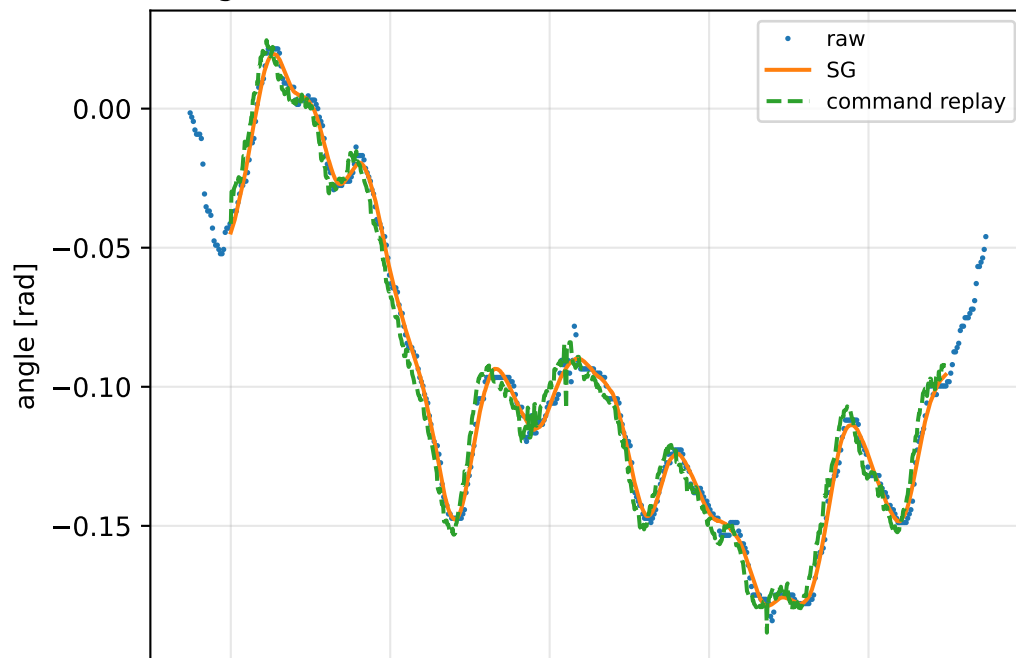


angular acceleration [rad/s²]

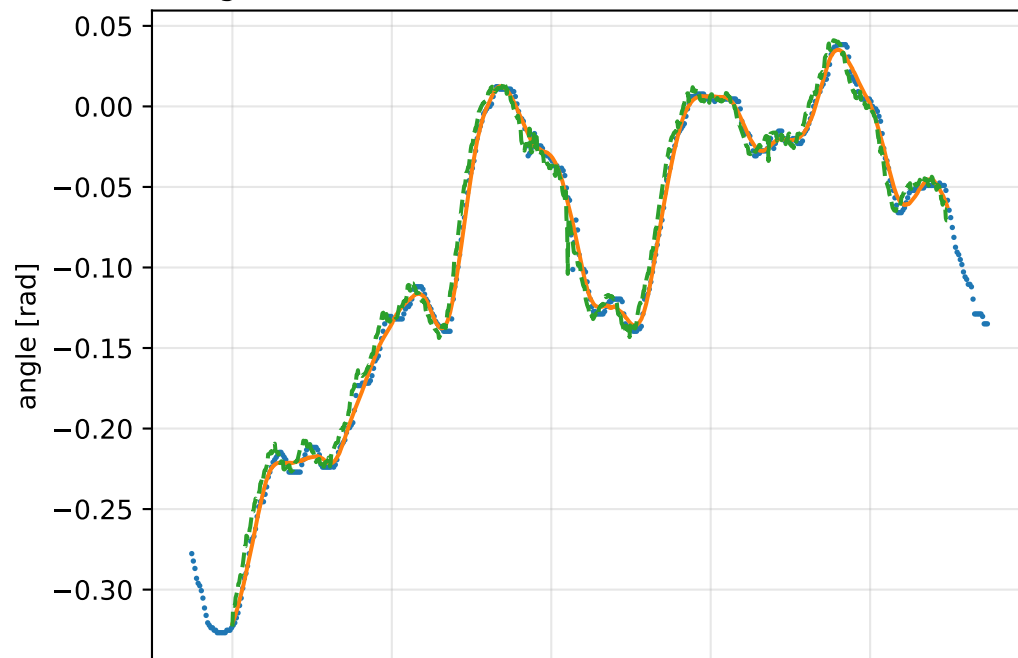


cov_full_no_R_uncertainty_in_s: gimbal measurement audit (objective=measured_sg)

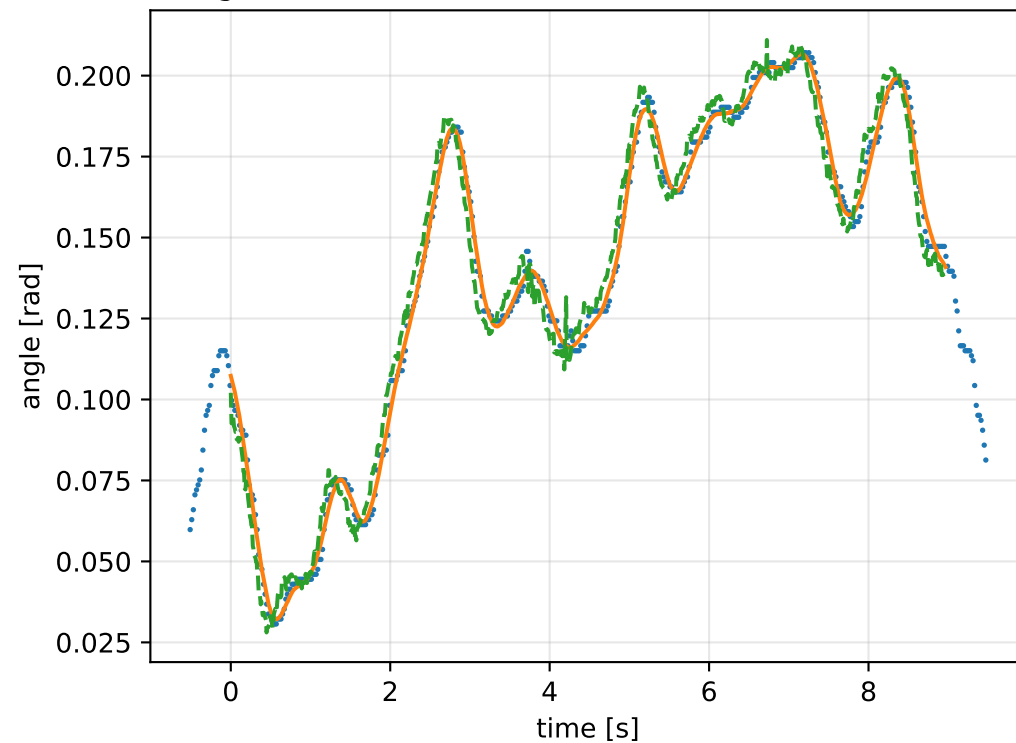
gimbal 1: RMSE=0.006345, max=0.01494 rad



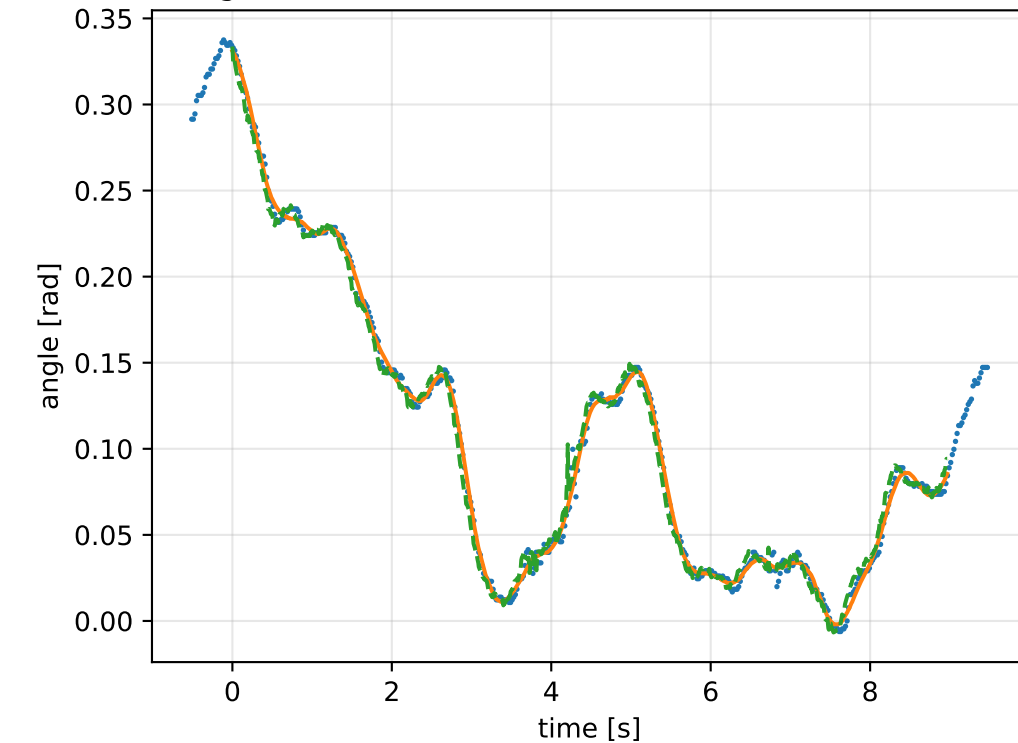
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

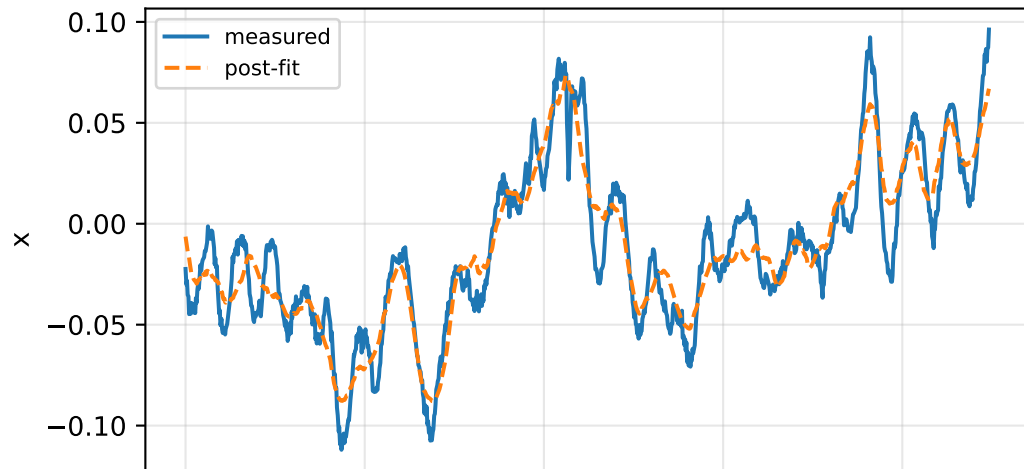


gimbal 4: RMSE=0.006321, max=0.03642 rad

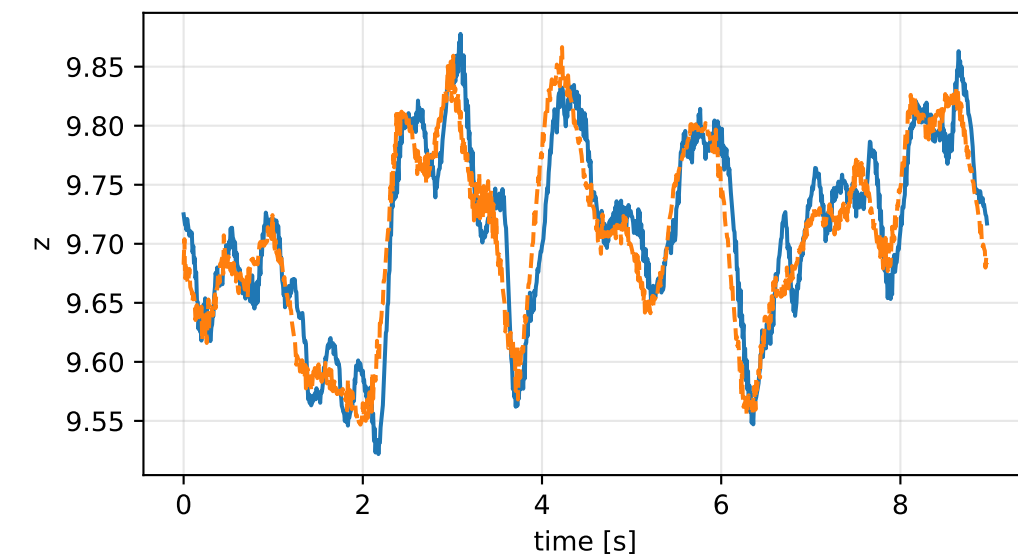
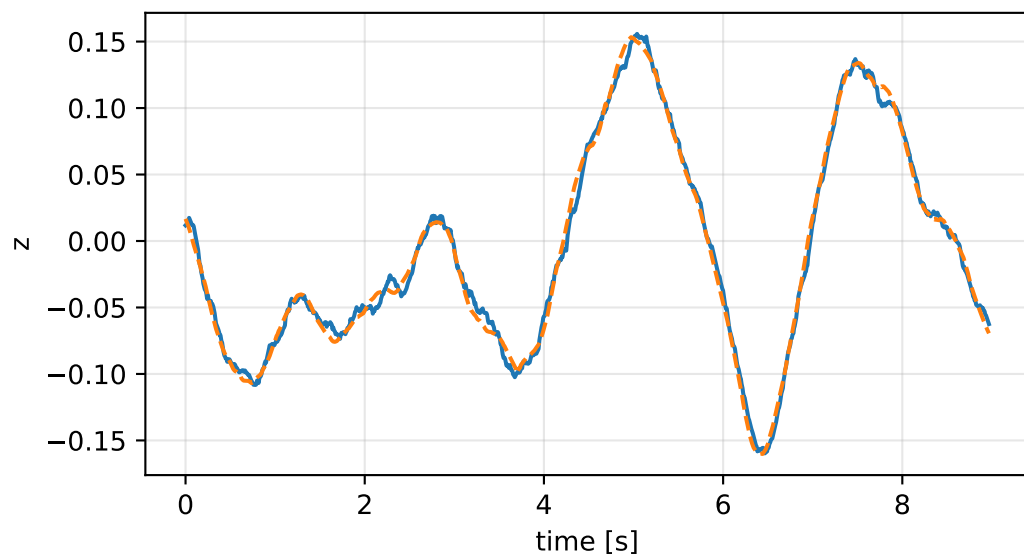
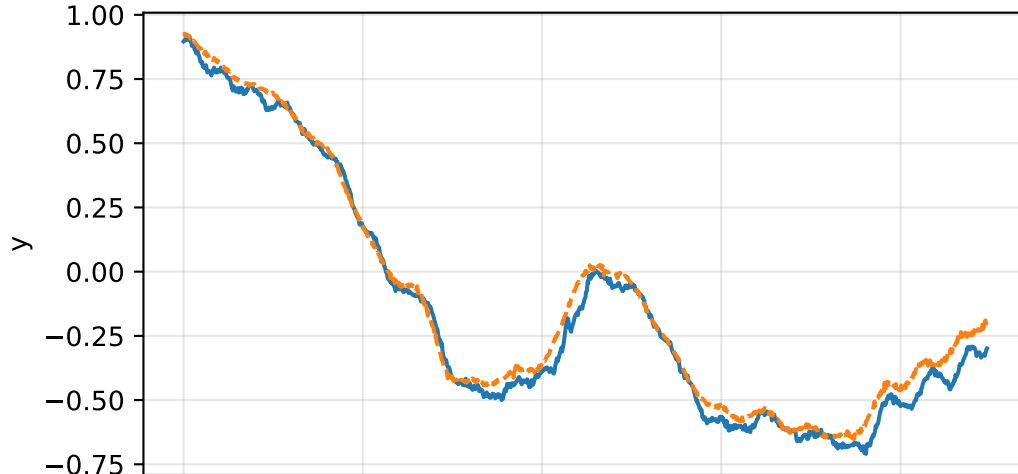
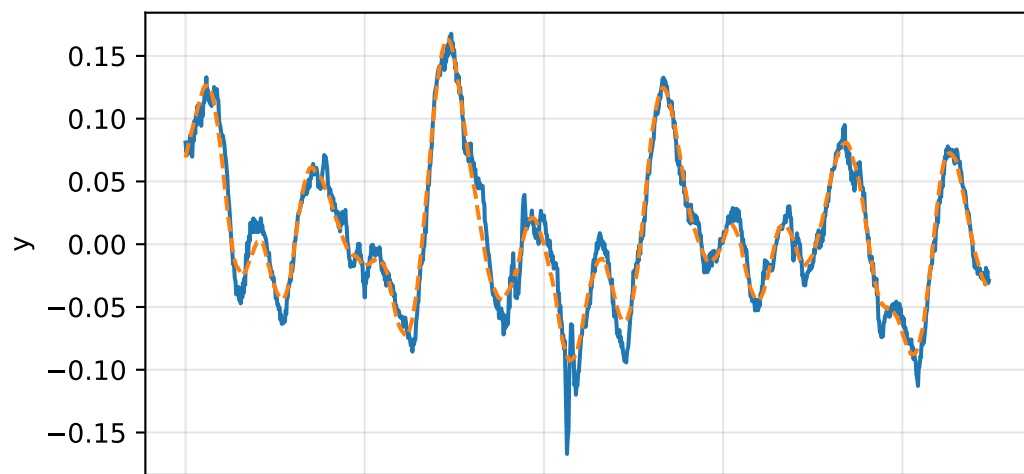
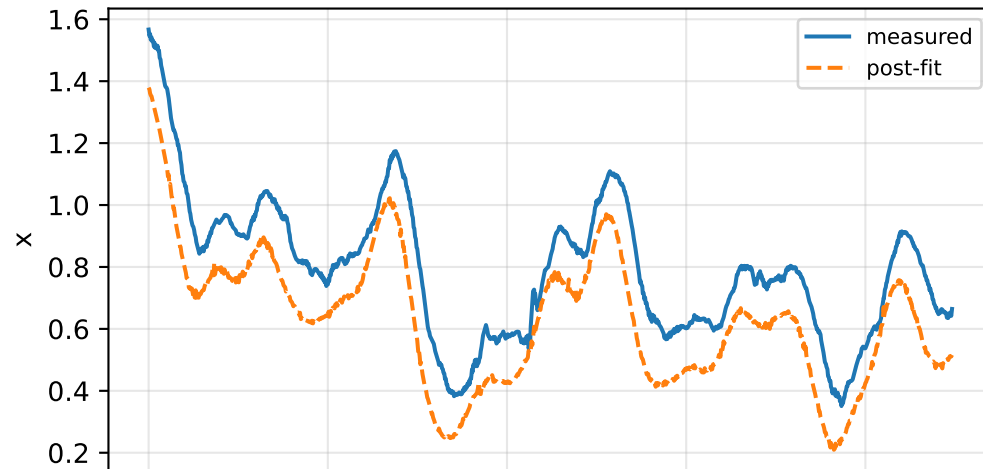


cov_full_no_R_uncertainty_in_s: IMU comparison (diagnostic only)

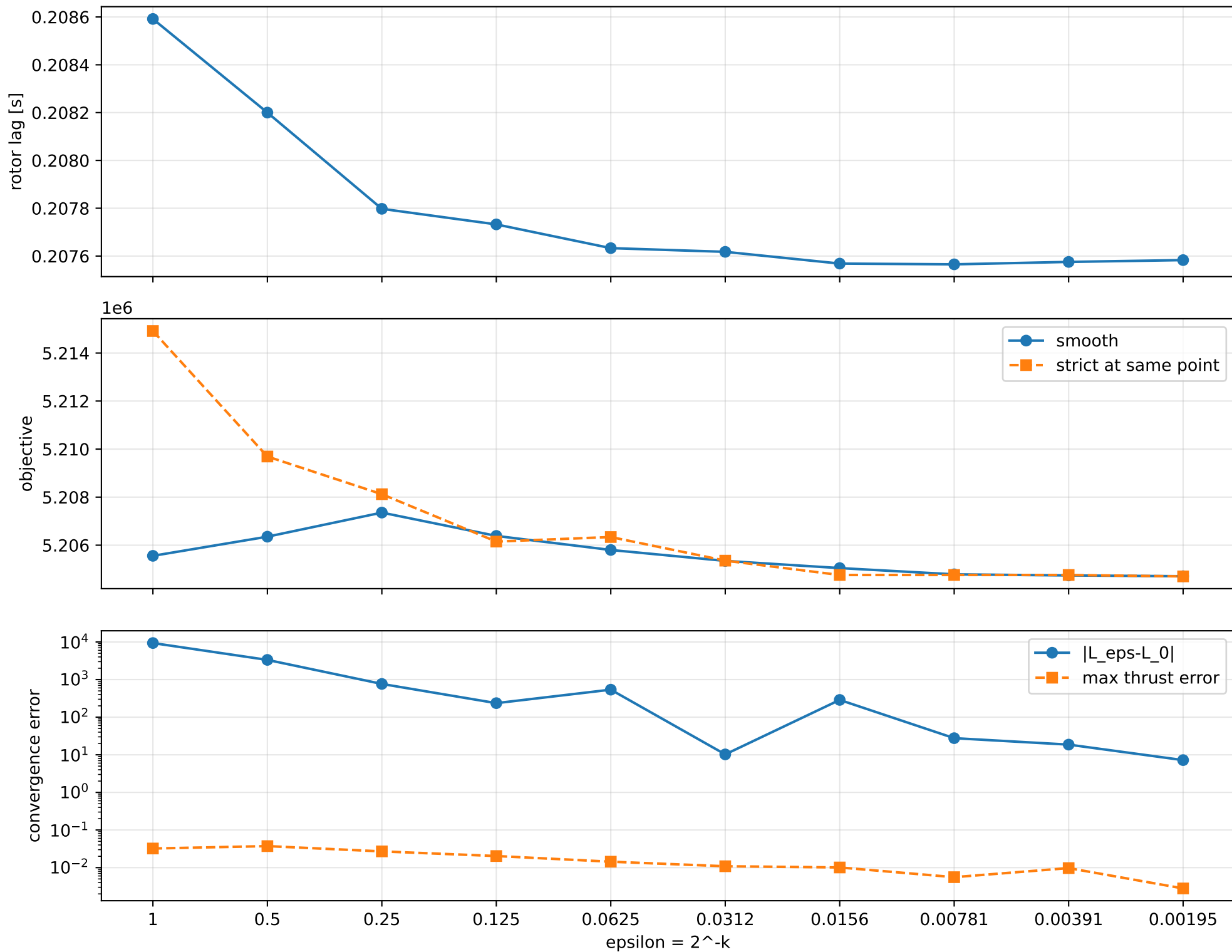
gyro [rad/s]



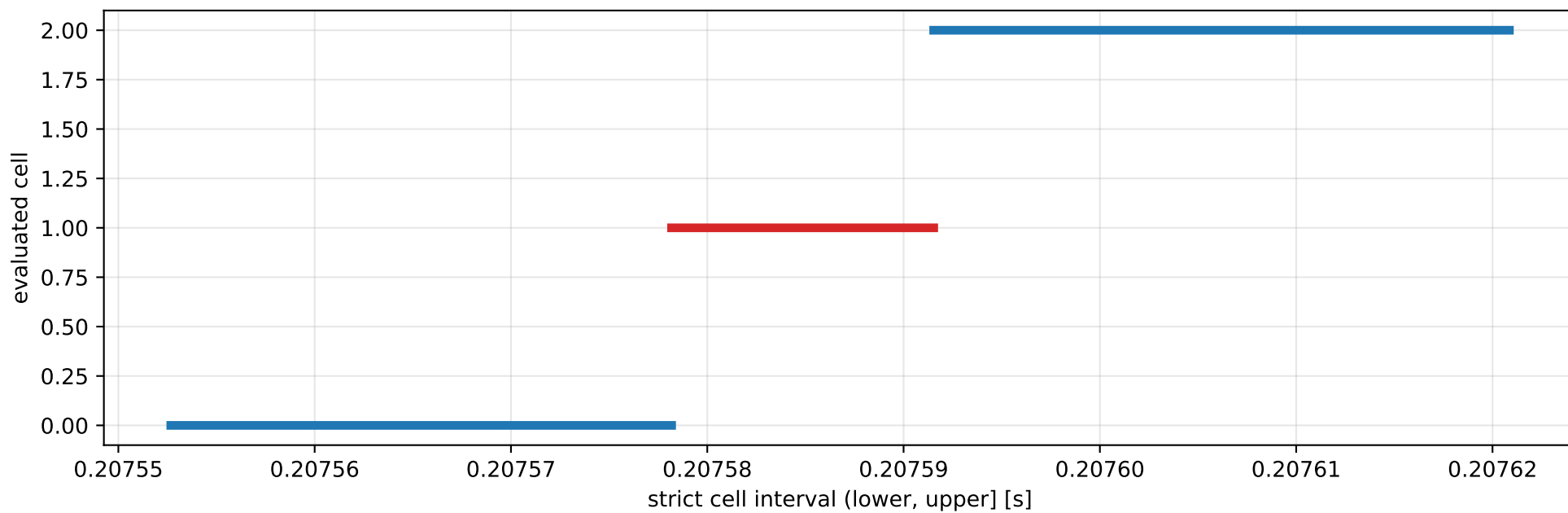
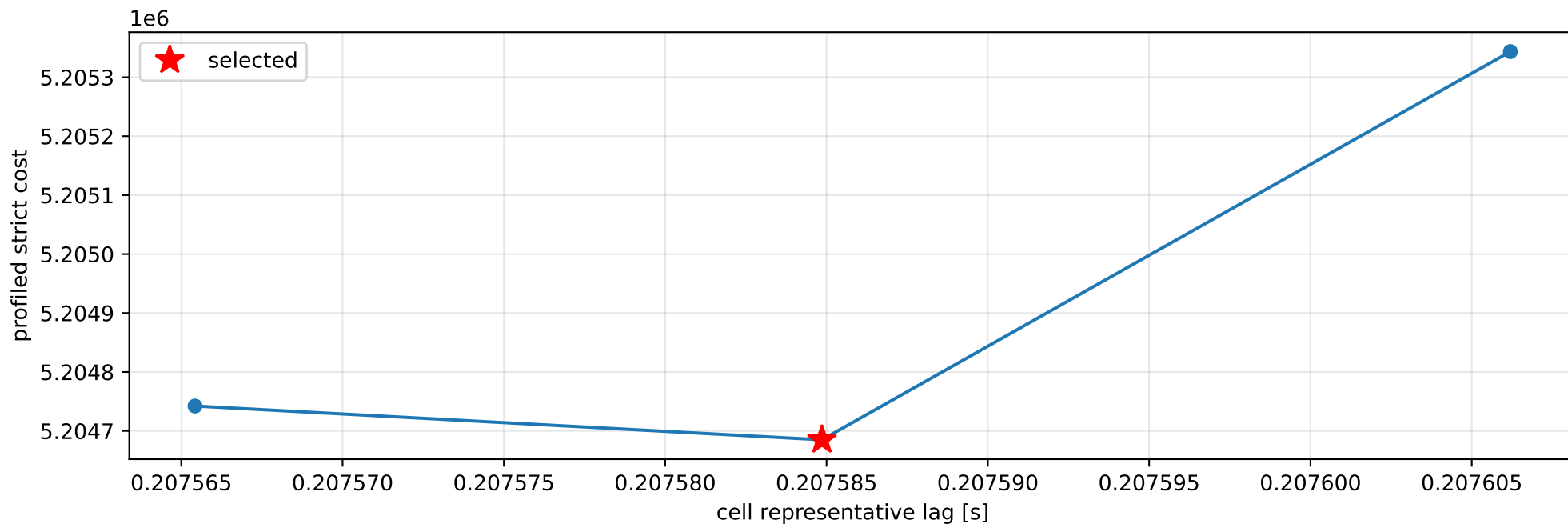
specific force [m/s^2]



cov_full_no_R_uncertainty_in_s: smooth-to-strict continuation



cov_full_no_R_uncertainty_in_s: exact strict-ZOH lag cells



selected (0.207578182, 0.207591534] s; final smooth 0.207583012 s; data boundary=False

cov_full_no_R_uncertainty_in_s: parameter estimate

Case: cov_full_no_R_uncertainty_in_s
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.22088431

inertia [kg m²]:

```
[[ 253.34604195 -96.56118719 -110.70116688]
 [ -96.56118719 229.11577782 -125.50010631]
 [-110.70116688 -125.50010631 194.77067717]]
```

CoG body [m]: [-1.10815375 -1.20890505 -1.4133801]

force effectiveness: [0.33988506 0.39144183 0.39706869 0.53723785]

scale-free J/m [m²]:

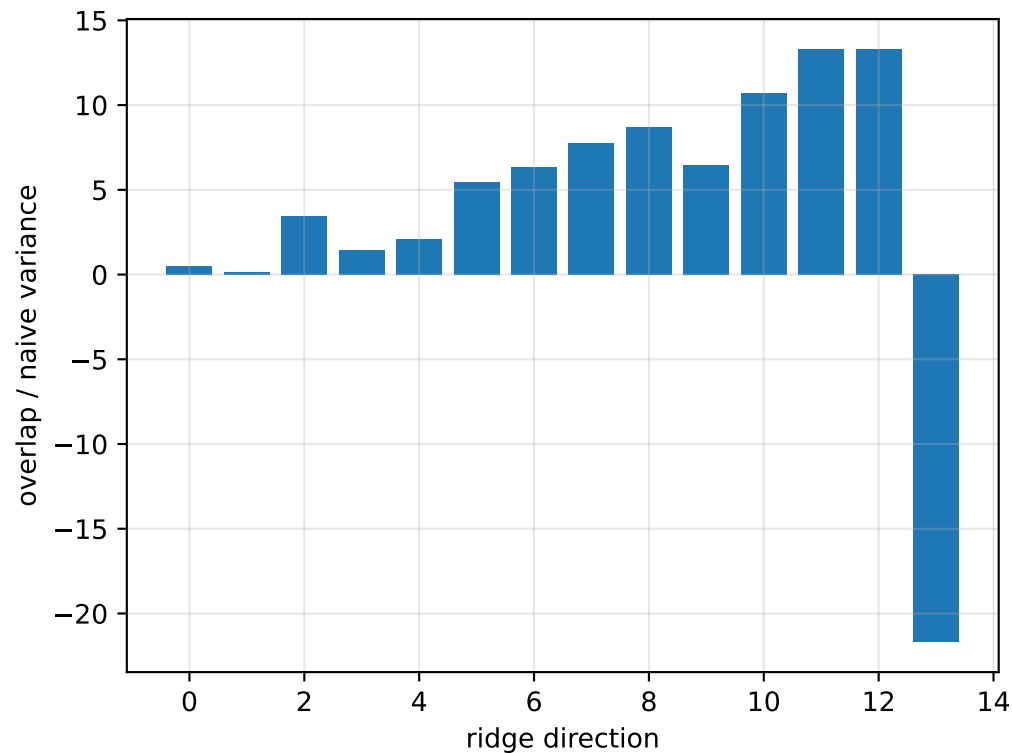
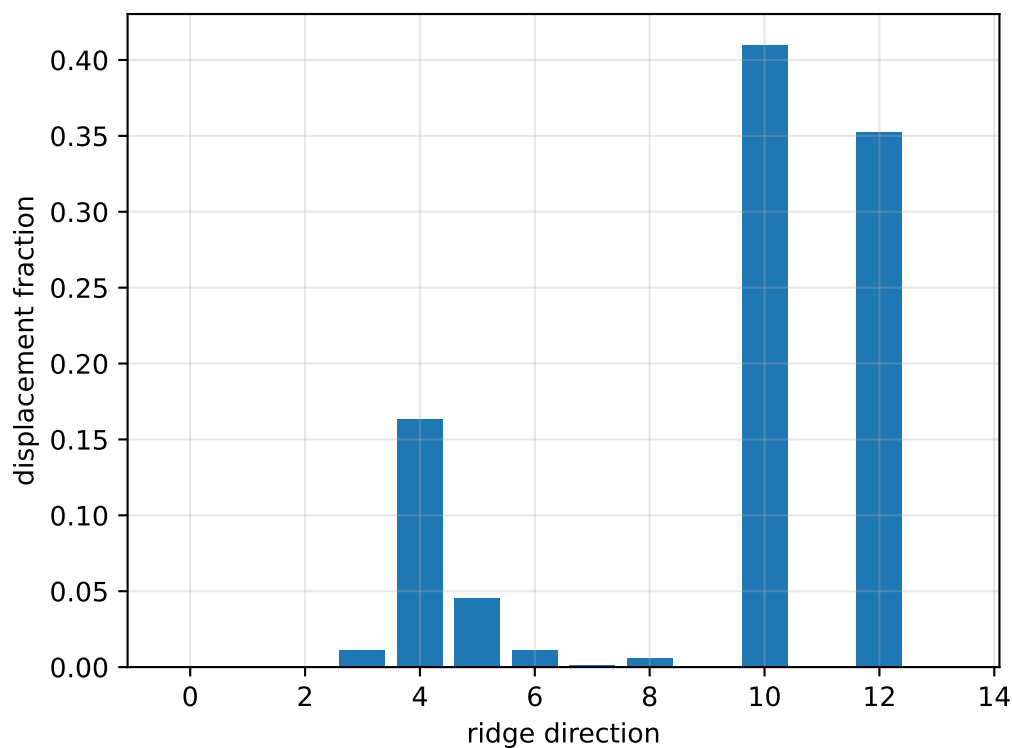
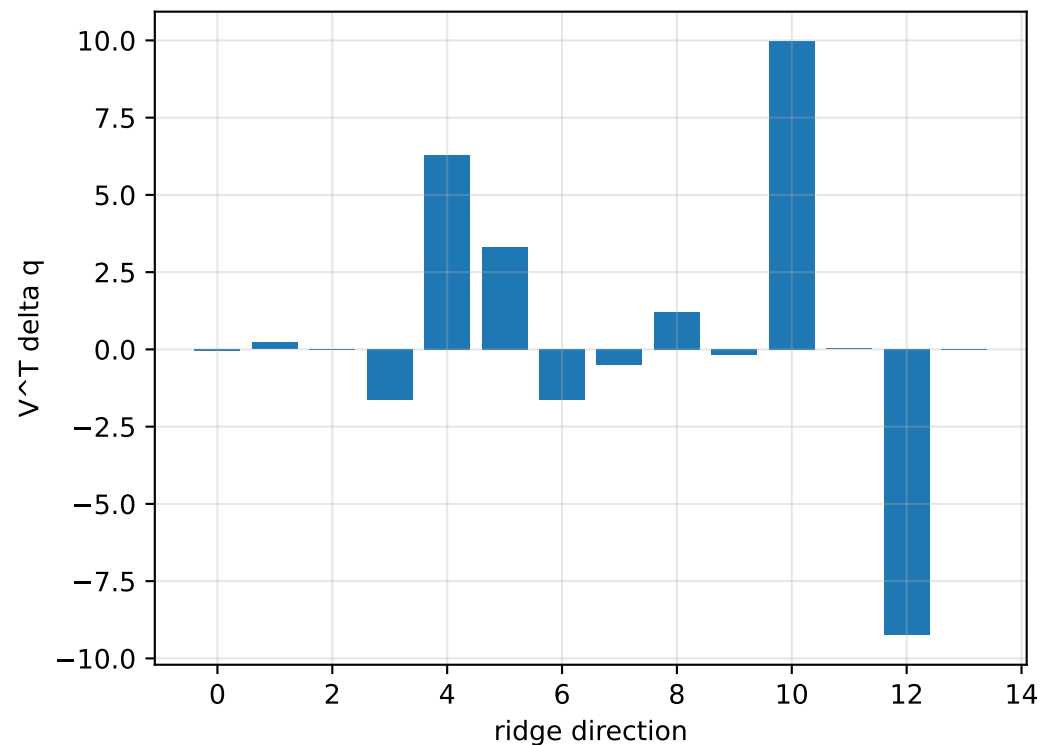
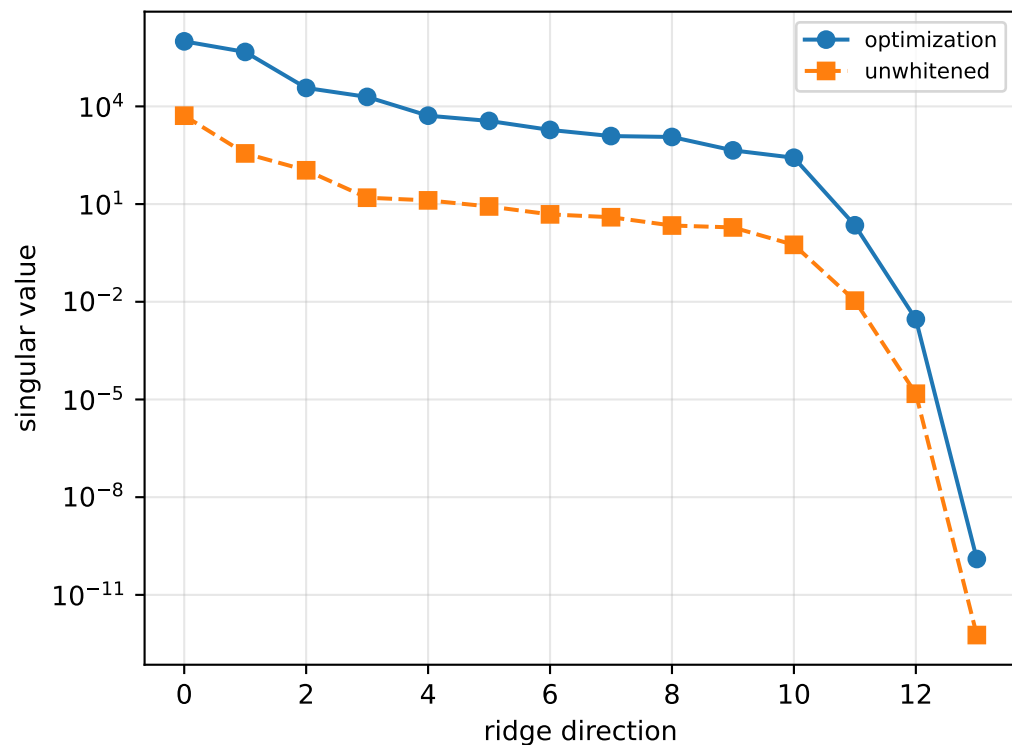
```
[[ 207.51027738 -79.09118526 -90.67293757]
 [ -79.09118526 187.66379077 -102.79442959]
 [ -90.67293757 -102.79442959 159.53245977]]
```

scale-free f/m: [0.27839252 0.32062156 0.3252304 0.44003993]

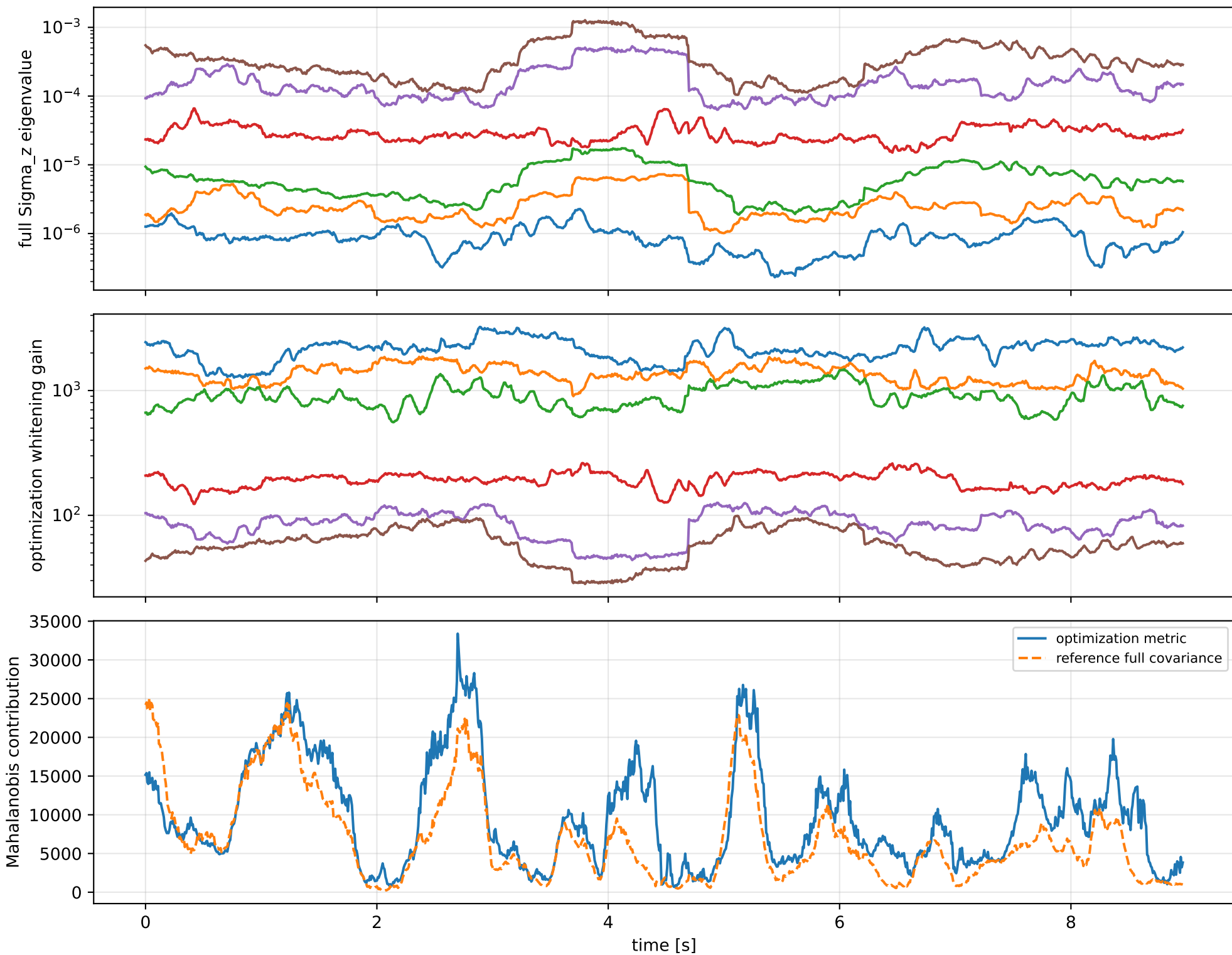
rotor lag cell: (0.207578182, 0.207591534) s

representative: 0.207584858 s

cov_full_no_R_uncertainty_in_s: ridge spectrum (rank 13, nullity 1, ||jv||=4.36e-10)



cov_full_no_R_uncertainty_in_s: optimization vs reference covariance



cov_full_no_R_uncertainty_in_s: wrench / closure (force max $3.66\text{e-}15$, torque max $6.39\text{e-}14$)

